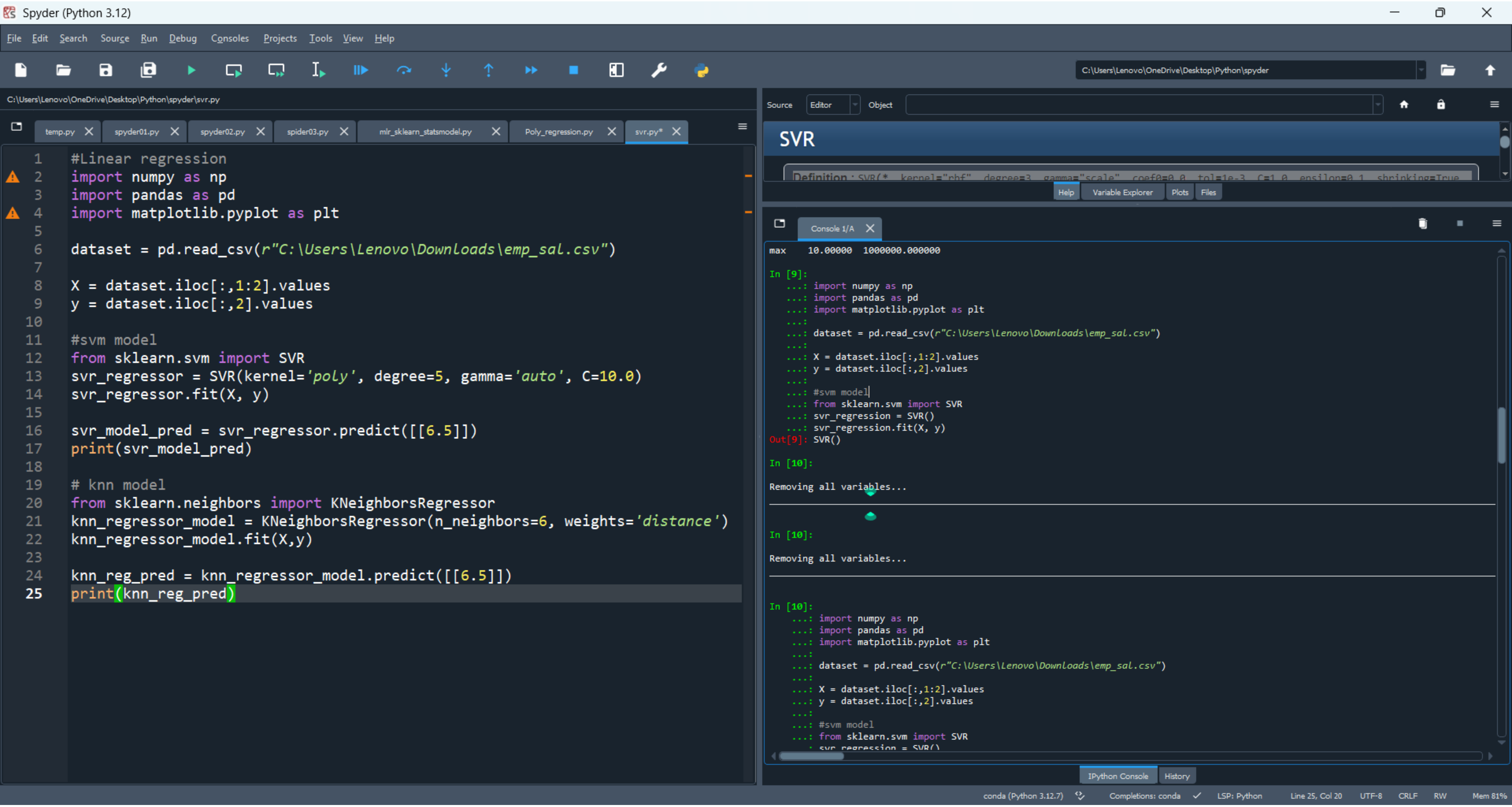


Spyder (Python 3.12)
File Edit Search Source Run Debug Consoles Projects Tools View Help
C:\Users\Lenovo\OneDrive\Desktop\Python\spyder
C:\Users\Lenovo\OneDrive\Desktop\Python\spyder\svr.py
temp.py
spyder01.py
spyder02.py
spider03.py
mlr_sklearn_statsmodel.py
Poly_regression.py
svr.py*
#Linear regression
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
dataset = pd.read_csv(r"C:\Users\Lenovo\Downloads\emp_sal.csv")
X = dataset.iloc[:,1:2].values
y = dataset.iloc[:,2].values
#svm model
from sklearn.svm import SVR
svr_regressor = SVR(kernel='poly', degree=5, gamma='auto', C=10.0)
svr_regressor.fit(X, y)
svr_model_pred = svr_regressor.predict([[6.5]])
print(svr_model_pred)
# knn model
from sklearn.neighbors import KNeighborsRegressor
knn_regressor_model = KNeighborsRegressor(n_neighbors=6, weights='distance')
knn_regressor_model.fit(X,y)
knn_reg_pred = knn_regressor_model.predict([[6.5]])
print(knn_reg_pred)
Source Editor Object
SVR
Help Variable Explorer Plots Files
Console 1/A
max 10.00000 1000000.000000
In [9]:
...: import numpy as np
...: import pandas as pd
...: import matplotlib.pyplot as plt
...: dataset = pd.read_csv(r"C:\Users\Lenovo\Downloads\emp_sal.csv")
...: X = dataset.iloc[:,1:2].values
...: y = dataset.iloc[:,2].values
...: #svm model
...: from sklearn.svm import SVR
...: svr_regression = SVR()
...: svr_regression.fit(X, y)
Out[9]: SVR()
In [10]:
Removing all variables...
In [10]:
Removing all variables...
In [10]:
...: import numpy as np
...: import pandas as pd
...: import matplotlib.pyplot as plt
...: dataset = pd.read_csv(r"C:\Users\Lenovo\Downloads\emp_sal.csv")
...: X = dataset.iloc[:,1:2].values
...: y = dataset.iloc[:,2].values
...: #svm model
...: from sklearn.svm import SVR
IPython Console History
conda (Python 3.12.7) Completions: conda LSP: Python Line 25, Col 20 UTF-8 CRLF RW Mem 81%

Spyder (Python 3.12)

File Edit Search Source Run Debug Consoles Projects Tools View Help

C:\Users\Lenovo\OneDrive\Desktop\Python\spyder\svr.py

temp.py | spyder01.py | spyder02.py | spider03.py | mlr_sklearn_statsmodel.py | Poly_regression.py | svr.py*

```python
#Linear regression
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt

dataset = pd.read_csv(r"C:\Users\Lenovo\Downloads\emp_sal.csv")

X = dataset.iloc[:,1:2].values
y = dataset.iloc[:,2].values

#svm model
from sklearn.svm import SVR
svr_regressor = SVR(kernel='poly', degree=5, gamma='auto', C=10.0)
svr_regressor.fit(X, y)

svr_model_pred = svr_regressor.predict([[6.5]])
print(svr_model_pred)

# knn model
from sklearn.neighbors import KNeighborsRegressor
knn_regressor_model = KNeighborsRegressor(n_neighbors=6, weights='distance')
knn_regressor_model.fit(X,y)

knn_reg_pred = knn_regressor_model.predict([[6.5]])
print(knn_reg_pred)
```

Source: Editor Object

SVR

Definition : SVR(*, kernel="rbf", degree=3, gamma="scale", coef0=0.0, tol=1e-3, C=1.0, epsilon=0.1, shrinking=True

Help | Variable Explorer | Plots | Files

Console 1/A

```
   ...: #svm model
   ...: from sklearn.svm import SVR
   ...: svr_regression = SVR()
   ...: svr_regression.fit(X, y)
Out[10]: SVR()

In [11]:
   ...: import numpy as np
   ...: import pandas as pd
   ...: import matplotlib.pyplot as plt
   ...:
   ...: dataset = pd.read_csv(r"C:\Users\Lenovo\Downloads\emp_sal.csv")
   ...:
   ...: X = dataset.iloc[:,1:2].values
   ...: y = dataset.iloc[:,2].values
   ...:
   ...: #svm model
   ...: from sklearn.svm import SVR
   ...: svr_regressor = SVR()
   ...: svr_regressor.fit(X, y)
Out[11]: SVR()

In [12]:
   ...: import numpy as np
   ...: import pandas as pd
   ...: import matplotlib.pyplot as plt
   ...:
   ...: dataset = pd.read_csv(r"C:\Users\Lenovo\Downloads\emp_sal.csv")
   ...:
   ...: X = dataset.iloc[:,1:2].values
   ...: y = dataset.iloc[:,2].values
   ...:
   ...: #svm model
   ...: from sklearn.svm import SVR
   ...: svr_regressor = SVR()
   ...: svr_regressor.fit(X, y)
   ...:
   ...: svr_model_pred = svr_regressor.predict([[6.5]])
   ...: print(svr_model_pred)
[130001.82883924]

In [13]:
   ...: from sklearn.svm import SVR
```

IPython Console | History

conda (Python 3.12.7) Completions: conda LSP: Python Line 25, Col 20 UTF-8 CRLF RW Mem 82%

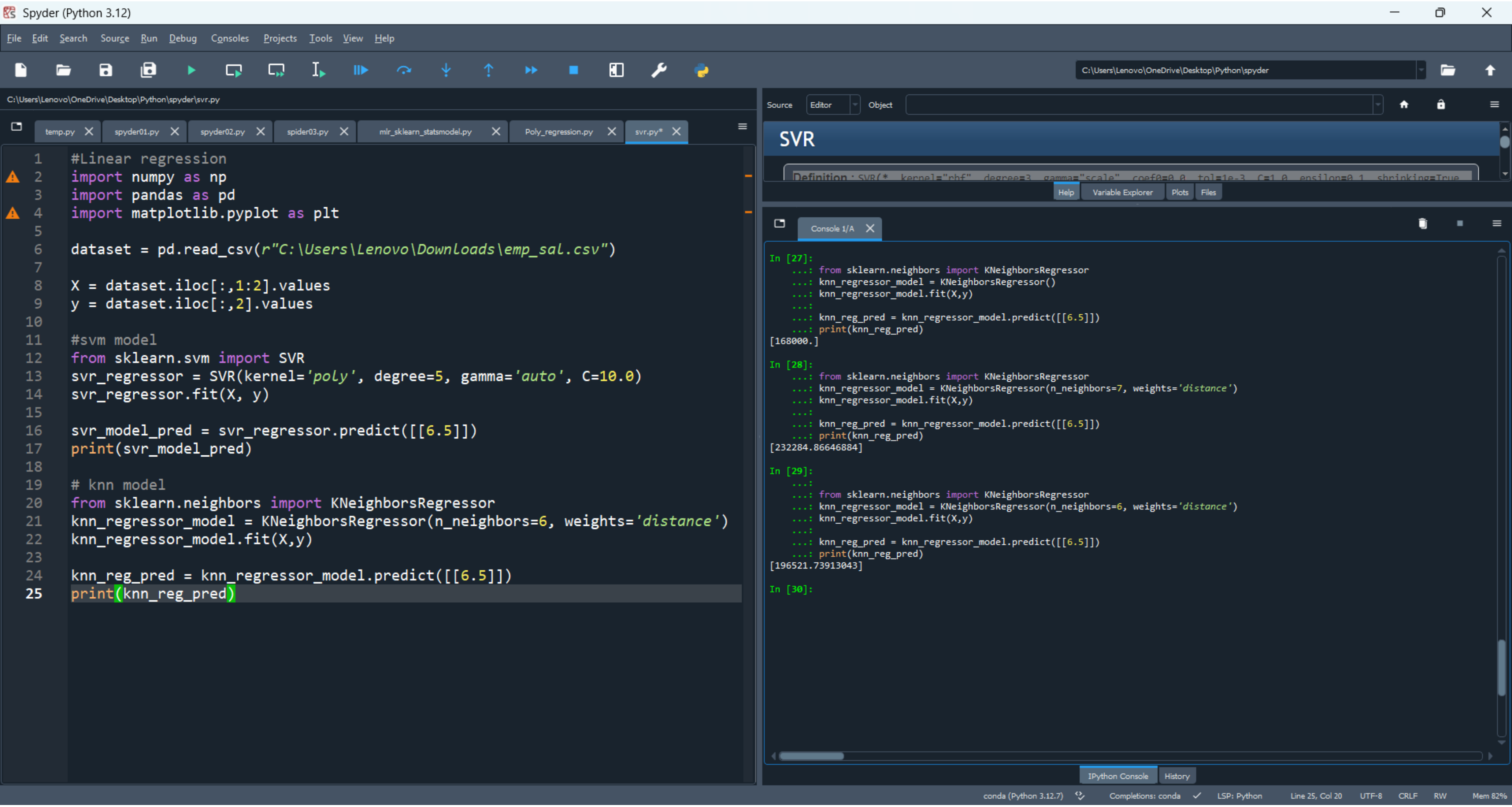
Spyder (Python 3.12)

C:\Users\Lenovo\OneDrive\Desktop\Python\spyder\svr.py

```python
#Linear regression
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt

dataset = pd.read_csv(r"C:\Users\Lenovo\Downloads\emp_sal.csv")

X = dataset.iloc[:,1:2].values
y = dataset.iloc[:,2].values

#svm model
from sklearn.svm import SVR
svr_regressor = SVR(kernel='poly', degree=5, gamma='auto', C=10.0)
svr_regressor.fit(X, y)

svr_model_pred = svr_regressor.predict([[6.5]])
print(svr_model_pred)

# knn model
from sklearn.neighbors import KNeighborsRegressor
knn_regressor_model = KNeighborsRegressor(n_neighbors=6, weights='distance')
knn_regressor_model.fit(X,y)

knn_reg_pred = knn_regressor_model.predict([[6.5]])
print(knn_reg_pred)
```

Console 1/A

```
In [13]:
    ...: from sklearn.svm import SVR
    ...: svr_regressor = SVR(kernel='sigmoid', degree=4, gamma='auto')
    ...: svr_regressor.fit(X, y)
Out[13]: SVR(degree=4, gamma='auto', kernel='sigmoid')

In [14]: svr_model_pred = svr_regressor.predict([[6.5]])
    ...: print(svr_model_pred)
[129999.99995299]

In [15]: from sklearn.svm import SVR
    ...: svr_regressor = SVR(kernel='poly', degree=4, gamma='scale')
    ...: svr_regressor.fit(X, y)
Out[15]: SVR(degree=4, kernel='poly')

In [16]: svr_model_pred = svr_regressor.predict([[6.5]])
    ...: print(svr_model_pred)
[134160.63306167]

In [17]:
    ...: from sklearn.svm import SVR
    ...: svr_regressor = SVR(kernel='poly', degree=5, gamma='scale')
    ...: svr_regressor.fit(X, y)
Out[17]: SVR(degree=5, kernel='poly')

In [18]: svr_model_pred = svr_regressor.predict([[6.5]])
    ...: print(svr_model_pred)
[164079.01344549]

In [19]:
    ...:
    ...: from sklearn.svm import SVR
    ...: svr_regressor = SVR(kernel='poly', degree=5, gamma='auto')
    ...: svr_regressor.fit(X, y)
Out[19]: SVR(degree=5, gamma='auto', kernel='poly')

In [20]: svr_model_pred = svr_regressor.predict([[6.5]])
    ...: print(svr_model_pred)
[159973.68854139]

In [21]:
    ...:
    ...: from sklearn.svm import SVR
```



Spyder (Python 3.12)
File Edit Search Source Run Debug Consoles Projects Tools View Help
C:\Users\Lenovo\OneDrive\Desktop\Python\spyder
C:\Users\Lenovo\OneDrive\Desktop\Python\spyder\svr.py
temp.py
spyder01.py
spyder02.py
spider03.py
mlr_sklearn_statsmodel.py
Poly_regression.py
svr.py*
#Linear regression
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
dataset = pd.read_csv(r"C:\Users\Lenovo\Downloads\emp_sal.csv")
X = dataset.iloc[:,1:2].values
y = dataset.iloc[:,2].values
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svr_regressor = SVR(kernel='poly', degree=5, gamma='auto', C=10.0)
svr_regressor.fit(X, y)
svr_model_pred = svr_regressor.predict([[6.5]])
print(svr_model_pred)
# knn model
from sklearn.neighbors import KNeighborsRegressor
knn_regressor_model = KNeighborsRegressor(n_neighbors=6, weights='distance')
knn_regressor_model.fit(X,y)
knn_reg_pred = knn_regressor_model.predict([[6.5]])
print(knn_reg_pred)
Source
Editor
Object
SVR
Help
Variable Explorer
Plots
Files
Console 1/A
In [27]:
...: from sklearn.neighbors import KNeighborsRegressor
...: knn_regressor_model = KNeighborsRegressor()
...: knn_regressor_model.fit(X,y)
...:
...: knn_reg_pred = knn_regressor_model.predict([[6.5]])
...: print(knn_reg_pred)
[168000.]
In [28]:
...: from sklearn.neighbors import KNeighborsRegressor
...: knn_regressor_model = KNeighborsRegressor(n_neighbors=7, weights='distance')
...: knn_regressor_model.fit(X,y)
...:
...: knn_reg_pred = knn_regressor_model.predict([[6.5]])
...: print(knn_reg_pred)
[232284.86646884]
In [29]:
...:
...: from sklearn.neighbors import KNeighborsRegressor
...: knn_regressor_model = KNeighborsRegressor(n_neighbors=6, weights='distance')
...: knn_regressor_model.fit(X,y)
...:
...: knn_reg_pred = knn_regressor_model.predict([[6.5]])
...: print(knn_reg_pred)
[196521.73913043]
In [30]:
IPython Console
History
conda (Python 3.12.7)
Completions: conda
LSP: Python
Line 25, Col 20
UTF-8
CRLF
RW
Mem 82%